

From A to B

You are given two integers, a and b . You want to transform a to b by performing a sequence of operations. You can only perform the following operations:

- Divide by two (only if a is even)
- Add one to a

What is the minimum number of these operations you need to transform a into b ?

Input

The single line of input contains two space-separated integers a and b ($1 \leq a, b \leq 10^9$). You want to transform a into b using only the operations given above.

Output

Output a single integer, which is the minimum number of the given operations needed to transform a into b .

Sample Input 1

103 27

Sample Output 1

4

Sample Input 2

3 8

Sample Output 2

5